



J.P. MORGAN PAYMENT CARTRIDGE FOR SALESFORCE COMMERCE CLOUD (SFRA)

Implementation Guide

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REVISION HISTORY

VERSION	DATE	AUTHOR	SUMMARY OF CHANGES
1.0	2026-03-05	JPMC	Initial Draft

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1. PURPOSE AND SCOPE

1.1 Purpose

This guide provides step-by-step instructions to install, configure, test, and deploy the J.P. Morgan Payment Cartridge for Salesforce Commerce Cloud using SFRA.

1.2 Scope

In scope

- Card payment processing (authorize, capture, refund, void)
- PIE encryption and SafeTech tokenization configuration
- Optional fraud check configuration
- Business Manager (CSC) payment operations enablement
- Testing and go-live readiness steps

2. OVERVIEW

The J.P. Morgan Payment Cartridge integrates J.P. Morgan's payment processing services with an SFCC SFRA storefront.

2.1 Key Capabilities

- **Card Payments:** authorize, capture, refund, void for Visa, Mastercard, American Express, Discover
- **Page Integrated Encryption (PIE):** encrypts card data in the browser before it reaches the server, reducing PCI scope
- **SafeTech Tokenization:** replaces PAN with tokens for secure storage and reuse
- **Fraud Check (Optional):** pre-authorization risk scoring to accept, review, or block transactions
- **Business Manager Extensions:** operational actions (capture/refund/void) from the CSC order view

2.2 Process Flow (High Level)

1. Shopper enters card details at checkout.
2. PIE scripts encrypt card data in the shopper's browser.
3. Cartridge submits encrypted payment payload for authorization.
4. J.P. Morgan returns authorization outcome and SafeTech token.
5. Capture occurs immediately or later depending on configured capture method.

3. SOLUTION COMPONENTS

Cartridge

Purpose

`int\_jpmc\_sfra`

SFRA storefront overlay
(checkout UI, hooks, client-side
scripts)

`int\_jpmc\_core`

Core payment logic (services,
authentication, helpers)

`bm\_jpmc`

Business Manager extensions
(CSC payment actions, tools)

4. PREREQUISITES AND REQUIREMENTS

4.1 Inputs Required from J.P. Morgan Payments Representative

- Merchant ID
- OAuth2 Client ID
- Token URI (OAuth2 token endpoint)
- Resource ID (OAuth2 resource identifier)
- PIE Merchant ID
- Certificate (if you provided a CSR)

4.2 SFCC Environment Requirements

- SFCC instance with **SFRA** installed
- Business Manager access (Administrator permissions recommended)
- `app\_storefront\_base` deployed
- Local workstation: Node.js and npm

5. INSTALLATION

5.1 Configure `dw.json`

Configure sandbox credentials in the root of the `link\_jpmc` folder.

> See Appendix A for a standardized sample.

5.2 Install Dependencies

```
```bash
npm install
```
```

5.3 Build Client-Side Assets

```
```bash
npm run build
```
```

5.4 Upload Cartridges

Upload the following cartridges (WebDAV / Prophet / equivalent):

- `int\_jpmc\_sfra`
- `int\_jpmc\_core`
- `bm\_jpmc`

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6. METADATA IMPORT

6.1 Purpose

Metadata import creates required services, site preferences, custom objects, and object extensions in Business Manager.

6.2 Prepare the Import Archive

1. Create a local folder named `jpmc\_metadata`.
2. Copy from `link\_jpmc/metadata/` into `jpmc\_metadata/`:
 - `Services.xml`
 - `system-objecttype-extensions.xml`
 - `custom-objecttype-definitions.xml`
3. Zip the folder as `jpmc\_metadata.zip`.

6.3 Upload and Import in Business Manager

1. **Administration > Site Development > Site Import & Export**
2. Under **Import**, click **Upload** and select `jpmc\_metadata.zip`.
3. Select the uploaded zip and click **Import**.
4. Confirm import completes with **SUCCESS**.

[Administration](#) > [Site Development](#) > Site Import & Export

Site Import & Export

This page allows you to export the current configuration of your organization including all of its sites. To download an archive, just click its file name.

Import

Upload Archive:

☒ Local ☐ Remote

No file chosen

6.4 Objects and Configurations Created

| File | Creates |
|-------------------------------------|--|
| `Services.xml` | HTTP services, credentials, service profile |
| `system-objecttype-extensions.xml` | Site preferences and groups, Order and Payment custom attributes |
| `custom-objecttype-definitions.xml` | `JPMCAccessToken` custom object for OAuth token + caching |

7. CARTRIDGE SETUP (BM CONFIGURATION)

7.1 Update Storefront Cartridge Path

1. **Administration** > **Sites** > **Manage Sites**
2. Select your storefront site.
3. **Settings** tab > **Cartridges**
4. Add `int\_jpmc\_sfra` and `int\_jpmc\_core` **before** `app\_storefront\_base`:

```
`int_jpmc_sfra:int_jpmc_core:app_storefront_base`
```

5. Click **Apply**.

[Administration](#) > [Sites](#) > [Manage Sites](#) > RefArch - Settings

General **Settings** Cache Site Status Page Meta Tag Rules

RefArch - Settings

Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

Instance Type:

Deprecated. The preferred way of configuring HTTP and HTTPS hostnames is by using new features of the site aliases configuration ("SEO > Aliases Configuration"). The HTTP/HTTPS hostname values set in t

HTTP Hostname:

HTTPS Hostname:

Instance Type: All

Cartridges:

7.2 Update Business Manager Cartridge Path

1. **Administration** > **Sites** > **Manage Sites**
2. Select **Business Manager**.
3. **Settings** tab > **Cartridges**
4. Add `bm\_jpmc` and `int\_jpmc\_core` **before** `bm\_app\_storefront\_base`:

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```
`bm_jpmc:int_jpmc_core:bm_app_storefront_base`
```

5. Click **Apply**.

[Administration](#) > [Sites](#) > [Manage Sites](#) > Business Manager - Settings

[Settings](#)
[Cache](#)
[Hostnames](#)

Business Manager - Settings

Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

Instance Type: Sandbox/Development

Deprecated. Up to two instance specific hostname aliases for Business Manager can be configured here.
 ⚠ Setting a different hostname here will make some Business Manager modules unreachable. Manage additional hostnames in the 'Hostnames' tab instead.

HTTP Hostname:

HTTPS Hostname:

Instance Type: All

Cartridges: bm_jpmc:int_jpmc_core:bm_app_storefront_base

7.3 Verify Services are Available

1. **Administration > Operations > Services**

2. Confirm these services appear:

| Service ID | Purpose |
|--------------------|-------------------------|
| JPMCAccessToken | OAuth2 token generation |
| JPMCPaymentService | Payment authorization |
| JPMCPaymentCapture | Payment capture |
| JPMCPaymentRefund | Payment refund |
| JPMCPaymentVoid | Payment void |
| JPMCFraudCheck | Fraud risk check |

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JPMCFraudRetrieve

Retrieve fraud results

JPMCFraudRetrieveById

Retrieve fraud result by ID

JPMCVerification

Card verification

8. CERTIFICATES AND KEYS

8.1 Purpose

The cartridge uses an X.509 certificate and private key to sign JWT tokens for API authentication.

8.2 Upload Certificate and Private Key

1. **Administration > Operations > Private Keys and Certificates**
2. Click **Import Key/Certificate**
3. Upload private key with alias: `jpmc-private-key`
4. Upload certificate with alias: `jpmc-certificate`

Important: Alias values must match what you configure in site preferences.

[Administration](#) / [Operations](#) /

Private Keys and Certificates [?](#)

| ☐ Alias | Hostnames | Type |
|---|-----------|---------------------|
| ☐ jpmc-certificate | | Trusted Certificate |
| ☐ jpmc-private-key | | Private Key |

9. KEY ID (KID) GENERATION – ONE TIME ACTIVITY

9.1 Purpose

The Key ID (`kid`) is a SHA-1 thumbprint of your certificate used in the JWT header.

9.2 Enable Thumbprint Generator in Business Manager

To See **JPMC Tools > Certificate Thumbprint Generator** in Merchant Tools

1. **Administration > Organization > Roles > Business Manager Modules**
2. Select Site (context)
3. Click **Certificate Thumbprint Generator**
4. Click Update

[Administration](#) > [Organization](#) > [Roles](#) > Administrator - Business Manager Modules

General Users **Business Manager Modules** Functional Permissions WebDAV Permissions Locale Permissions Price Adjustment Limits Customer Service Center Permissions

Administrator - Business Manager Modules

This list shows all Business Manager modules available in the system for which permissions can be granted. Click **Select Context** to select the context for which you would like to modify permissions. Available context options are the organization, one site, or multiple sites. Select the checkboxes and use the **Update** button at the bottom of the page to grant permissions to certain Business Manager modules. Deselect the checkboxes and use the **Update** button to revoke permissions to specific Business Manager modules. Read access may be granted to an increasing number of Business Manager modules. Write access includes read access. When removing read access, make sure you haven't also granted write access for the respective feature. When multiple contexts are selected with different permissions for a feature or module, detailed information can be viewed by clicking in the "Details" column.

Selected Context: RefArch
[Select Context](#)

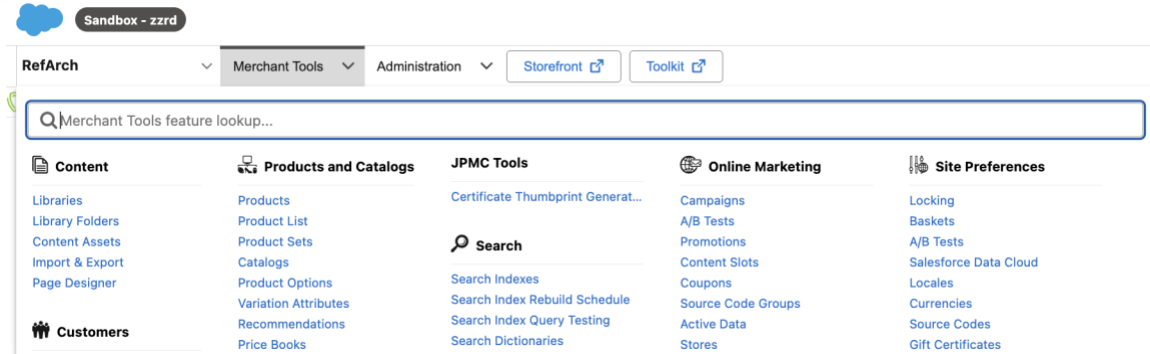
| Business Manager Module | Module Description | Read | Write |
|----------------------------------|---|-------------------------------------|-------------------------------------|
| JPMC Tools | | ☐ | ☐ |
| Certificate Thumbprint Generator | Generate SHA-1 thumbprint from BM Certificate Manager | ☒ | ☒ |

9.3 Generate Thumbprint in Business Manager

1. **Merchant Tools > JPMC Tools > Certificate Thumbprint Generator**
2. Enter certificate alias (default: `jpmc-certificate`)
3. Click **Generate Thumbprint**

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4. Click Save to Site Preferences to Save this value in Site Preference



10. SITE PREFERENCES CONFIGURATION

10.1 Open Preference Groups

Merchant Tools > Site Preferences > Custom Preferences

Configure both groups:

- **JPMC Payment Configuration**
- **JPMC Certificate & JWT Settings**

10.2 JPMC Payment Configuration

| PREFERENCE | DESCRIPTION |
|-------------------------|-----------------------------|
| JPMC Client ID | OAuth2 Client ID |
| JPMC Merchant Id | Merchant identifier |
| JPMC Tokenization Type | Token type |
| JPMC Capture Method | Capture timing |
| Enable JPMC Fraud Check | Pre-auth fraud scoring |
| JPMC Token URI | OAuth2 endpoint |
| JPMC Resource ID | OAuth2 resource |
| JPMC Audience | JWT audience |
| JPMC Token Expiration | JWT lifetime |
| JPMC PIE Get Key URL | `getkey.js` URL with PIE ID |

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| | |
|--------------------------------|------------------------------|
| JPMC PIE Encryption URL | `encryption.js` URL |
| JPMC Merchant Software Company | Header value |
| JPMC Merchant Software Product | Header value |
| JPMC Merchant Software Version | Header value |
| JPMC Platform ID | Optional platform identifier |

Click **Save**.

10.2.1 Capture Method Options

| VALUE | BEHAVIOR |
|---------|---|
| MANUAL | Authorization only; capture later via CSC |
| DELAYED | Authorization with delayed auto-capture (120-minute window) |
| NOW | Immediate sale (authorize + capture in one request) |

10.3 JPMC Certificate & JWT Settings

| PREFERENCE | DESCRIPTION | DEFAULT |
|------------|-------------|---------|
|------------|-------------|---------|

| | | |
|------------------------|---------------------------------|------------------|
| JPMC Certificate Alias | Uploaded certificate alias | jpmc-certificate |
| JPMC Private Key Alias | Uploaded private key alias | jpmc-private-key |
| JPMC Key ID (kid) | SHA-1 thumbprint from Section 9 | NA |

Click **Save**.

11. PAYMENT METHODS SETUP

11.1 Enable Credit Card Payment Method

1. **Merchant Tools > Ordering > Payment Methods**
2. Locate `CREDIT\_CARD` (Has to be enabled)
3. Ensure **Enabled**
4. Set **Payment Processor** to `JPMC\_Payment`
5. Apply changes

[Merchant Tools > Ordering > Payment Methods](#)

Payment Methods

| Payment Methods | | | |
|--|---------------|---------|------------|
| To learn about Salesforce Payments, see here . | | | |
| Payment methods are managed here. To create a new payment method, click the New button. To remove a payment method click the remove icon in the payment method row. The default payment methods can't be removed, and their IDs can't be changed. When you select the CREDIT_CARD payment method, credit/debit cards can be reordered through drag-and-drop | | | |
| New Sort Order Credit/Debit Cards Import/Export | | | |
| ID | Name | Enabled | Sort Order |
| AdyenComponent | Adyen Payment | Yes | 7 |
| BANK_TRANSFER | Bank Transfer | No | 8 |
| BMC | Bit Mo Later | No | 12 |
| CREDIT_CARD | Credit Card | Yes | 10 |

11.2 Verify Accepted Card Types

| CARD TYPE | CARD BRAND |
|------------|------------------|
| Visa | Visa |
| Mastercard | Mastercard |
| Amex | American Express |
| Discover | Discover |

11.3 Future Payment Methods

Apple Pay and Google Pay support is planned for a future release.

12. FRAUD CHECK CONFIGURATION

12.1 Enable Fraud Check

1. **Merchant Tools > Site Preferences > Custom Preferences**
2. Open **JPMC Payment Configuration**
3. Set **Enable JPMC Fraud Check** = `true`
4. Click **Save**

12.2 Configure Risk Thresholds

Set **JPMC Fraud Risk Thresholds** with your risk ranges.

> See **Appendix B** for a standardized sample.

| ACTION | BEHAVIOR |
|--------|---------------------------------|
| ACCEPT | Proceed to authorization |
| REVIEW | Proceed; flag for manual review |
| BLOCK | Decline before authorization |

13. ORDER MANAGEMENT IN BUSINESS MANAGER (CSC)

13.1 Access JPMC Payments Tab

1. **Merchant Tools > Ordering > Orders**
2. Open an order
3. Select **JPMC Payments** tab

Order No.: 00008201
Creation Date: 3/3/2026 11:51 pm
Source Code:

Order Status: Status: OPEN
Confirmation Status: NOTCONFIRMED
Export Status: NOTEXPORTED
Shipping Status: NOTSHIPPED
Payment Status: NOTPAID
Is A Gift: ☐

Billing Address: env vars test
env vars test
United States 77056
Houston TX
env-vars@test.com

Payment Method: CREDIT_CARD Visa [More >](#)
[Details... \(1\)](#) **JPMC Payments**

SHIPMENT 1
Ship To: env vars test
env vars test
United States 77056
Houston TX

| NAME | AVAILABILITY | QUANTITY | PRICE | TAX | TOTAL |
|---|----------------|----------|----------|---------|----------|
| Robert Ludlum's: The Bourne Conspiracy (for X-Box 360)
sierra-the-bourne-conspiracy-xbox360M | Available (89) | 1 | \$ 59.99 | \$ 3.00 | \$ 59.99 |

Shipping Method: Ground
Tax: \$ 0.30

[Add Note](#)

ITEMS TOTAL: **\$ 59.99**
SHIPPING TOTAL: \$ 5.99
TAX TOTAL: \$ 3.30
ORDER TOTAL: **\$ 69.28**

13.2 Available Actions

| ACTION | DESCRIPTION | WHEN AVAILABLE |
|---------|--|--------------------------------------|
| Capture | Capture authorized funds (full amount) | After authorization (MANUAL/DELAYED) |
| Refund | Refund captured payment (full/partial) | After capture |
| Void | Cancel authorization before capture | After authorization, before capture |

Note: If capture method is `NOW`, capture occurs at authorization and **Capture** is not shown.

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Order Status:
Status: OPEN
Information Status: NOTCONFIRMED
Port Status: NOTEXPORTED
Shipping Status: NOTSHIPPED
Payment Status: NOTPAID
Gift: ☺

Game: e Bourne Conspiracy (for X-Box 360)
SKU: nspracy-xbox360M

JPMC Payments

JPMC Payment Management

| | | | |
|----------------|--------------------------------------|----------------|--------------------------|
| Order Number | 00008201 | Payment Method | Credit Card |
| Order Total | USD 69.28 | Payment Status | Authorized & Captured |
| Transaction ID | 99b225e5-6232-4725-8571-1aa5aa2e5e4d | Card | Visa *****1111 |
| Capture Method | NOW (SALE) | Auth Timestamp | 2026-03-03T18:20:46.633Z |
| Authorized | | Captured | Available for Capture |
| USD 69.28 | | USD 69.28 | Available for Refund |
| | | USD 0.00 | USD 69.28 |

Payment Actions

Refund

Refund Amount: Max: 69.28 ☐ Refund Full Amount

[Refund](#)

14. TESTING AND VALIDATION

14.1 Test Card Payments

Use J.P. Morgan sandbox test cards per J.P. Morgan developer documentation.

<https://developer.payments.jpmorgan.com/docs/commerce/online-payments/testing>

14.2 Test Scenarios

| SCENARIO | EXPECTED RESULT |
|--------------------------|----------------------------------|
| Successful authorization | Order placed; payment authorized |
| Capture (MANUAL) | Funds captured via CSC |
| Refund after capture | Refund processed via CSC |
| Void before capture | Authorization voided via CSC |
| Declined card | Order not placed; error shown |
| Fraud check — ACCEPT | Proceeds normally |
| Fraud check — REVIEW | Proceeds; flagged in logs |
| Fraud check — BLOCK | Declined pre-authorization |

14.3 Verify Service Communication Logs

1. **Administration > Operations > Services**

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2. Select a JPMC service
3. Confirm **Communication Log Enabled** = `true`
4. **Administration > Site Development > Development Setup > Log Files**
5. Review logs prefixed `service-jpmc-`

15. GO-LIVE CHECKLIST

15.1 Credentials and Configuration

- Obtain production credentials (Client ID, Merchant ID, Token URI, Resource ID)
- Obtain production PIE URLs
- Upload production certificate and private key
- Generate production `kid` thumbprint
- Update site preferences with production values

Note: Private keys must be protected and access limited.

15.2 Services

- Update credential URLs from sandbox to production endpoints
- **Administration > Operations > Services > Credentials** tab
- Update each JPMC credential URL per J.P. Morgan production details

15.3 Payment Methods

- Confirm `CREDIT\_CARD` enabled with `JPMC\_Payment` processor
- Confirm accepted card types align to target market

15.4 Business Manager Paths

- Storefront cartridge path: `int\_jpmc\_sfra:int\_jpmc\_core:app\_storefront\_base`
- BM cartridge path: `bm\_jpmc:int\_jpmc\_core:bm\_app\_storefront\_base`

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15.5 Testing

- Complete production end-to-end payment
- Validate capture/refund/void in CSC
- Validate fraud thresholds (if enabled)
- Review communication logs

15.6 Security

- Confirm PIE encryption (no plaintext card data)
- Confirm CSRF protection on checkout forms

16. TROUBLESHOOTING

16.1 Authorization Fails

| POSSIBLE CAUSE | RESOLUTION |
|------------------------------------|--|
| Invalid/expired OAuth2 credentials | Validate Client ID, Token URI, Resource ID, Audience in site preferences |
| Certificate mismatch | Re-upload cert/key; regenerate `kid` |
| Incorrect service URLs | Verify in **Services > Credentials** |
| Service disabled | Ensure JPMC services are enabled |

16.2 PIE Encryption Not Working

| POSSIBLE CAUSE | RESOLUTION |
|--------------------|---|
| Incorrect PIE URLs | Validate Get Key URL and Encryption URL in site preferences |
| JavaScript errors | Check browser console for PIE errors |

16.3 Capture/Refund/Void Fails in CSC

| POSSIBLE CAUSE | RESOLUTION |
|-------------------------------------|---|
| Missing BM cartridge path | Confirm
`bm_jpmc:int_jpmc_core` is present |
| Transaction already captured/voided | Check payment transaction state |
| Service timeout | Review **Services > Profiles**
(default 30,000 ms) |

16.5 Viewing Logs

| LOG TYPE | BUSINESS MANAGER LOCATION |
|----------------------------|---|
| Custom error logs | Log Files: `customerror-*` |
| Service communication logs | Log Files: `service-jpmc-*` |
| Custom info/warn logs | Log Files: `custominfo-*` /
`customwarn-*` |

17. SUPPORT AND ESCALATION

- J.P. Morgan API issues: contact your J.P. Morgan technical account representative.
- Cartridge installation / SFCC platform issues: Contact your SFCC system integrator.

APPENDICES

APPENDIX A. SAMPLE 'DW.JSON'

```
{
  "hostname": "your-sandbox.demandware.net",
  "username": "your-bm-username",
  "password": "your-bm-password",
  "code-version": "version1"
}
```

Note: Do not commit secrets

APPENDIX B. SAMPLE FRAUD RISK THRESHOLDS JSON

```
{
  "low": {"min": 0, "max": 49, "action": "ACCEPT", "log": "info"},
  "medium": {"min": 50, "max": 69, "action": "REVIEW", "log": "warn"},
  "high": {"min": 70, "max": 100, "action": "BLOCK", "log": "warn"}
}
```

APPENDIX C. BUILD AND UPLOAD COMMANDS

```
```bash
```

npm install
-------------

npm run build
---------------

```
```
```